

## Appendix A. Hierarchy of the Champion & Seth (1968) Forest Classification

This appendix describes the hierarchical structure of the forest type classification of Champion & Seth (1968), drawn entirely from the authors' own text. All descriptions, terminology, and examples are from *A Revised Survey of the Forest Types of India* (Champion & Seth, 1968, Manager of Publications, Government of India, Delhi).

### A.1 Overview of the hierarchy

The classification has five nested levels. Each level subdivides the one above it on a different criterion:

| Level | Label              | Count | Basis of division                                                    |
|-------|--------------------|-------|----------------------------------------------------------------------|
| 1     | Major Group        | 6     | Broad thermal and moisture regime                                    |
| 2     | Group (Type Group) | 16    | Rainfall, altitude and moisture conditions within the Major Group    |
| 3     | Sub-group          | ~30   | Geographic region (lettered A, B, C, ...)                            |
| 4     | Type               | ~200  | Ecological status (C, E, S, DS, etc.) and dominant species           |
| 5     | Sub-type           | —     | Finer variation in rainfall, composition, or locality (a, b, c, ...) |

### A.2 Major Groups

The authors divide Indian forests into six Major Groups based on broad temperature and moisture regime. Temperature is treated through four zones — Tropical, Subtropical, Temperate, and Alpine — defined by mean annual temperature and mean January temperature. Within the Tropical zone, the authors make a primary subdivision into Moist and Dry, because moisture conditions during the growing season are "evidently the most important item in determining the type of vegetation which can exist."

| No. | Major Group                  | Temperature and moisture basis                                                       |
|-----|------------------------------|--------------------------------------------------------------------------------------|
| I   | Moist Tropical Forests       | Tropical zone (mean annual temp. >24°C, no frost); higher rainfall                   |
| II  | Dry Tropical Forests         | Tropical zone; lower rainfall, longer dry season                                     |
| III | Montane Sub-tropical Forests | Sub-tropical zone (mean annual 17–24°C, cool winter, frost rare); mainly hill tracts |
| IV  | Montane Temperate Forests    | Temperate zone (mean annual 7–17°C, pronounced winter with frost and snow)           |

| No. | Major Group        | Temperature and moisture basis                                                       |
|-----|--------------------|--------------------------------------------------------------------------------------|
| V   | Sub-alpine Forests | Transition from temperate to alpine; stunted forest near the timber limit            |
| VI  | Alpine Scrub       | Alpine zone (mean annual <7°C, severe winter with much snow); above the timber limit |

### A.3 Groups (Type Groups)

Within each Major Group, the authors recognise Groups (also called Type Groups) based on the character and appearance of the forest. For the Tropical Major Groups, the primary axis is moisture — the authors describe a series "of successively drier climatic types." For the Montane groups, altitude, rainfall during the growing season, and the balance between conifers and broadleaved species are the chief criteria.

The 16 Groups and the authors' own diagnostic descriptions are:

#### I — Moist Tropical Forests

**Group 1 — Tropical Wet Evergreen Forests.** "Dense tall forest, entirely evergreen or nearly so." Examples: West Coast tropical evergreen forest (1A/C4), Andamans tropical evergreen forest (1A/C2), Assam Valley tropical wet evergreen forest (1B/C1).

**Group 2 — Tropical Semi-Evergreen Forests.** "Dominants include deciduous species but evergreens predominate." Examples: Andamans semi-evergreen forest (2A/C1), West Coast semi-evergreen forest (2A/C2), Assam Valley semi-evergreen forest (2B/C1).

**Group 3 — Tropical Moist Deciduous Forests.** "Dominants mainly deciduous but sub-dominants and lower storeys largely evergreen. Top canopy rarely dense and even, but over 35 m. high." Examples: Very moist teak forest (3B/C1a), Moist sal-bearing forest (3C/C2), Moist plains sal forest (3C/C2d).

**Group 4 — Littoral and Swamp Forests.** "Mainly evergreen, of varying density and height, but always associated predominantly with wetness." The authors subdivide this Group into five Sub-groups: Littoral forest (on coastal sand), Tidal Swamp forest (on estuarine mud), Freshwater Swamp forest (on wet alluvium), Seasonal Swamp forest (on alluvium subject to seasonal submersion), and Riverain Fringing forest (along stream sides where subsoil water is available throughout the year). Examples: Littoral forest (4A/L1), Mangrove forest (4B/TS2), *Myristica* swamp forest (4C/FS1).

#### II — Dry Tropical Forests

**Group 5 — Tropical Dry Deciduous Forests.** "Entirely deciduous or nearly so. Top canopy rather light and rarely over 25 m. high, usually 8–20 m." Examples: Dry teak-bearing forest (5A/C1), Dry

sal-bearing forest (5B/C1), Northern dry mixed deciduous forest (5B/C2), Southern dry mixed deciduous forest (5A/C3).

**Group 6 — Tropical Thorn Forests.** "Deciduous with low thorny trees and xerophytes predominating. Canopy more or less broken. Height under 10 m." Examples: Southern thorn forest (6A/C1), Desert thorn forest (6B/C1), Desert dune scrub (6/1S1).

**Group 7 — Tropical Dry Evergreen Forests.** "Hard leaved evergreen trees predominate with some deciduous emergents, often dense, but usually under 20 m. high." Example: Tropical dry evergreen forest (7/C1).

### **III — Montane Sub-tropical Forests**

**Group 8 — Sub-tropical Broadleaved Hill Forests.** "Broadleaved largely evergreen high forest." Examples: Nilgiri subtropical hill forest (8A/C1), East Himalayan subtropical wet hill forest (8B/C1).

**Group 9 — Sub-tropical Pine Forests.** "Pine associations predominate." Examples: Himalayan subtropical pine forest (9/C1), Assam subtropical pine forest (9/C1b).

**Group 10 — Sub-tropical Dry Evergreen Forests.** "Low xerophytic forest and scrub." Example: Subtropical dry evergreen forest (10/C1).

### **IV — Montane Temperate Forests**

**Group 11 — Montane Wet Temperate Forests.** "Evergreen forest without conifers." Examples: Southern montane wet temperate forest (11A/C1), East Himalayan wet temperate forest (11B/C1).

**Group 12 — Himalayan Moist Temperate Forests.** "Evergreen forest, mainly sclerophyllous oaks and conifers." Examples: Ban oak forest (12/C1), Moist deodar forest (12/C1a), Western mixed coniferous forest (12/C1b), Moist temperate deciduous forest (12/C2).

**Group 13 — Himalayan Dry Temperate Forests.** "Open coniferous forest with sparse xerophytic undergrowth." Examples: Dry deodar forest (13/C1), Neoza pine forest (13/C2), Dry temperate coniferous forest (13/C3).

### **V — Sub-alpine Forests**

**Group 14 — Sub-alpine Forests.** "Stunted deciduous or evergreen forest usually in close formation with or without conifers." Examples: West Himalayan sub-alpine birch/fir forest (14/C1), East Himalayan sub-alpine birch/fir forest (14/C2).

### **VI — Alpine Scrub**

**Group 15 — Moist Alpine Scrub.** "Low but often dense scrub." Examples: Birch/Rhododendron scrub forest (15/C1), Deciduous alpine scrub (15/C2), Dwarf Rhododendron scrub (15/C3).

**Group 16 — Dry Alpine Scrub.** "Xerophytic shrubs in open formation." Example: Dry alpine scrub (16/C1).

#### A.4 Sub-groups

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Within each Group, the authors separate geographic regions as lettered Sub-groups (A, B, C, ...). The basis is regional locality — "a forester in Odisha and a forester in Assam are managing different working circles" — rather than a floristic or physiognomic difference alone.

Not all Groups have Sub-groups. Groups 7, 9, 10, and 13 are not subdivided into lettered regions; their Types are listed directly under the Group number.

#### Example — Group 3 (Tropical Moist Deciduous Forests):

| Sub-group | Name                                 | Region                                                 |
|-----------|--------------------------------------|--------------------------------------------------------|
| 3A        | Andamans moist deciduous forests     | Andaman Islands                                        |
| 3B        | South Indian moist deciduous forests | Peninsular India south of the Satpuras                 |
| 3C        | North Indian moist deciduous forests | Gangetic plain, Himalayan foothills, and eastern India |
| 3D        | —                                    | (Not used; some editions list it, merged into 3C)      |

#### Example — Group 5 (Tropical Dry Deciduous Forests):

| Sub-group | Name                                    | Region                        |
|-----------|-----------------------------------------|-------------------------------|
| 5A        | Southern tropical dry deciduous forests | Peninsular India              |
| 5B        | Northern tropical dry deciduous forests | Northern India, Siwalik hills |

#### A.5 Types and the status codes

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Within a Sub-group, the authors list individual Types. Each Type carries a letter code that records its ecological status — whether it is the mature forest the climate supports, a soil-controlled variant, a transient stage, or a condition produced by human interference. The codes are:

| Code | Status          |
|------|-----------------|
| C    | Climatic climax |

| Code | Status            |
|------|-------------------|
| E    | Edaphic climax    |
| 1S   | Primary seral     |
| 2S   | Secondary seral   |
| DS   | Degradation stage |
| L    | Littoral          |
| TS   | Tidal swamp       |
| FS   | Freshwater swamp  |
| SS   | Seasonal swamp    |
| RS   | Riparian          |

The numeral following the letter is a serial index (C1, C2, C3 ...), not a rank.

### Definitions of the ecological status terms

The following definitions explain each status term in plain language. They are drawn from the ecological literature that Champion & Seth built upon — principally Clements (1916) and Tansley (1935) — and are consistent with how the authors used each term throughout the classification.

**Succession** is the natural process by which the plant community at a site changes over time, one community replacing another, moving step by step toward a stable endpoint. Champion & Seth describe it as communities developing "progressively towards a form which is in equilibrium with the prevailing climate" (following Clements 1916).

**Sere** is the full chain of communities that appear one after another at a site, from the first colonisers to the final stable community. Each community in this chain is called a **seral stage** — a temporary community that will eventually be replaced by the next one if left undisturbed (Clements 1916). Champion & Seth code seral types as S, signalling to the forest manager that the present vegetation is not the vegetation the site will end up carrying.

**Primary sere** is a sere that starts on bare ground that has never had vegetation before — a new riverbank, a sand dune, a landslide, a glacial deposit. There is no soil, no seed bank, no leftover roots — everything must build from scratch (Clements 1916). Champion & Seth code this **1S**. Examples: Khair–sissu forest growing on fresh river alluvium (5/1S2), Desert dune scrub colonising wind-blown sand (6/1S1).

**Secondary sere** is a sere that starts where a previous community was destroyed — by fire, clearing, or abandonment — but the soil, seeds, and root stocks are still in the ground. Recovery is faster than in a primary sere because the biological foundation is already there (Clements 1916). Champion & Seth code this **2S**. Examples: Secondary dry deciduous forest regrowing after destruction of the parent community (5/2S1), Pioneer Euphorbiaceous scrub recolonising cleared wet evergreen forest

(1/2S1).

**Climax** is a plant community that has reached a stable, self-maintaining state. It can reproduce itself generation after generation without outside help — if a tree dies, its replacement comes from within the same community. It is the endpoint of succession (Clements 1916). Champion & Seth do not commit to whether there is one climax per climate (monoclimax) or several (polyclimax), but their scheme names multiple stable endpoints within one climatic region, which is polyclimactic in practice.

**Climatic climax** is the stable forest community that the regional climate supports on ordinary, well-drained sites with soils of medium depth and fertility. Champion & Seth define it as "the type that predominates on sites and soils of medium depth and fertility." Coded **C**. If you leave such a site alone indefinitely, this is the forest it will grow and keep. Example: Dry sal-bearing forest (5B/C1) on normal upland sites in northern India.

**Edaphic climax** is a stable forest community that differs from the climatic climax of its region because the soil holds it there permanently. The climate could support something else, but unusual soil conditions — waterlogging, salinity, laterite, shallow rocky ground, heavy clay — prevent that from happening, permanently. Champion & Seth code this **E**. The key distinction: an edaphic type is stable like a climatic climax — it will not change on its own — which is what separates E from S in their system. Example: *Anogeissus pendula* forest (5/E1) on quartzite and shallow rocky ground — the climate could support dry sal forest here, but the soil cannot.

**Seral stage** is any of the intermediate, temporary communities in a sere. A seral community is one that is still changing — it has not yet reached its endpoint. If left alone, it will be replaced by the next community in the sequence. Champion & Seth code seral types as **1S** (primary) and **2S** (secondary) to signal that the present vegetation is transient.

**Degradation stage** is a forest that has been reduced in height, density, or species composition by sustained human pressure — overgrazing, repeated burning, lopping for fodder or fuel, unregulated felling. Champion & Seth describe these as forests "degraded to varying degrees through biotic influences." Coded **DS**. The authors regard them as recoverable: "The process will be reversed if protection is extended to degraded types." In other words, remove the pressure and the forest will grow back. This corresponds to what Tansley (1935) called a **plagioclimax** — a community held in a lowered state by ongoing interference. Example: Dry deciduous scrub (5/DS1) — what remains of dry deciduous forest after prolonged grazing and cutting.

Some Types are listed under a Sub-group heading and belong to that region. Others are listed under headings such as "General edaphic and seral types of dry deciduous forests" and belong to the Group as a whole — their controlling factor is substrate or disturbance rather than region.

**Example — Types within Sub-group 5B (Northern tropical dry deciduous forests):**

| Code   | Name                                | Status          |
|--------|-------------------------------------|-----------------|
| 5B/C1  | Dry sal-bearing forest              | Climatic climax |
| 5B/C1a | Dry Siwalik sal forest              | Sub-type of C1  |
| 5B/C1b | Dry plains sal forest               | Sub-type of C1  |
| 5B/C1c | Dry peninsular sal forest           | Sub-type of C1  |
| 5B/C2  | Northern dry mixed deciduous forest | Climatic climax |

**Example — General types listed at Group 5 level (not confined to one Sub-group):**

| Code  | Name                           | Status            |
|-------|--------------------------------|-------------------|
| 5/E1  | Anogeissus pendula forest      | Edaphic           |
| 5/E3  | Babul forest                   | Edaphic           |
| 5/E5  | Butea forest                   | Edaphic           |
| 5/E7  | Laterite thorn forest          | Edaphic           |
| 5/E8  | Saline/alkaline scrub savannah | Edaphic           |
| 5/1S1 | Dry tropical riverain forest   | Primary seral     |
| 5/1S2 | Khair–sissu forest             | Primary seral     |
| 5/1S3 | Inundation babul forest        | Primary seral     |
| 5/2S1 | Secondary dry deciduous forest | Secondary seral   |
| 5/DS1 | Dry deciduous scrub            | Degradation stage |
| 5/DS2 | Dry savannah forest            | Degradation stage |
| 5/DS3 | Euphorbia scrub                | Degradation stage |
| 5/DS4 | Dry grassland                  | Degradation stage |

## A.6 Sub-types

Where a Type shows further variation in rainfall, composition, or locality, the authors divide it into Sub-types lettered a, b, c, ...

**Example — Sub-types of 3B/C1 (Moist teak-bearing forest):**

| Code   | Name                       | Distinguishing feature   |
|--------|----------------------------|--------------------------|
| 3B/C1a | Very moist teak forest     | Wettest end of the range |
| 3B/C1b | Moist teak forest          | Intermediate moisture    |
| 3B/C1c | Slightly moist teak forest | Driest end of the range  |

**Example — Sub-types of 3C/C2 (Moist sal-bearing forest):**

| Code   | Name                        | Distinguishing feature  |
|--------|-----------------------------|-------------------------|
| 3C/C2a | Moist Siwalik sal forest    | Siwalik hills           |
| 3C/C2b | Moist bhabar sal forest     | Bhabar tract, foothills |
| 3C/C2c | Moist tarai sal forest      | Tarai wetlands          |
| 3C/C2d | Moist plains sal forest     | Gangetic plain          |
| 3C/C2e | Moist peninsular sal forest | Peninsula               |

## A.7 Reading the notation

A forest type code is read as:

(Group)(Sub-group) / (Type)(Sub-type)

**Worked example — 3C/C2e:**

| Part | Value       | Meaning                                              |
|------|-------------|------------------------------------------------------|
| 3    | Group 3     | Tropical Moist Deciduous Forests                     |
| C    | Sub-group C | North Indian moist deciduous forests                 |
| C2   | Type C2     | Climatic climax, 2nd type (Moist sal-bearing forest) |
| e    | Sub-type e  | Moist peninsular sal forest                          |

**Worked example — 5/1S2:**

| Part | Value        | Meaning                                      |
|------|--------------|----------------------------------------------|
| 5    | Group 5      | Tropical Dry Deciduous Forests               |
| —    | No Sub-group | General type, not confined to one region     |
| 1S2  | Type 1S2     | Primary seral, 2nd type (Khair–sissu forest) |

**Worked example — 4B/TS2:**

| Part | Value       | Meaning                                 |
|------|-------------|-----------------------------------------|
| 4    | Group 4     | Littoral and Swamp Forests              |
| B    | Sub-group B | Tidal swamp forests                     |
| TS2  | Type TS2    | Tidal swamp, 2nd type (Mangrove forest) |

## A.8 Floristic notation used in type descriptions

Each forest type in Champion & Seth is accompanied by a floristic listing that records the species present in the stand along with their stratum, leaf habit, and abundance. Three sets of codes appear in these listings. None is formally defined in the book; the definitions below are drawn from the ecological literature that the authors built upon.

### A.8.1 Species Groups (strata)

Champion & Seth organise the species of each type into vertical layers of the forest, which they call Species Groups and number with Roman numerals. The concept of forest stratification — the recognition that a forest is not a single layer but a stack of distinct communities at different heights — was established by Schimper (1903) and developed for tropical forests by Richards (1952), whose A, B, C storey terminology corresponds directly to Champion & Seth's I, II, III.

| Code | Stratum             | Definition                                                                                                                                                                                                                             |
|------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I    | Top canopy trees    | The uppermost continuous layer of the forest; the dominant trees that form the main canopy, receiving full sunlight. Richards (1952) calls this the A storey.                                                                          |
| II   | Second storey trees | A sub-canopy layer of smaller trees growing beneath the main canopy, shade-tolerant or waiting to replace canopy trees. Richards (1952) calls this the B storey.                                                                       |
| Ila  | Bamboos             | Bamboos are separated from the second storey because their growth form, phenology, and ecological behaviour differ fundamentally from broadleaved trees; they frequently form dense gregarious patches that suppress other vegetation. |
| III  | Shrubs              | Woody plants of low stature forming the understorey beneath both canopy layers. Richards (1952) calls this the C storey.                                                                                                               |
| IVa  | Herbs               | Non-woody flowering plants of the ground layer, excluding grasses.                                                                                                                                                                     |
| IVb  | Grasses             | Graminoids of the ground layer, listed separately from herbs because their response to light, fire, and grazing differs from that of broadleaved herbs and they are ecologically diagnostic of open or disturbed conditions.           |

| Code | Stratum   | Definition                                                                                                                                                                                                        |
|------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V    | Climbers  | Woody or herbaceous plants that use other plants for mechanical support; lianas and rattan canes are the most conspicuous in Indian tropical forests.                                                             |
| Epi. | Epiphytes | Plants that grow on other plants for physical support but derive no nutrition from the host — orchids, ferns, mosses, lichens. Their presence and abundance are indicators of canopy continuity and air humidity. |
| Par. | Parasites | Plants that attach to a host and derive water and/or nutrients from it — <i>Loranthus</i> , <i>Viscum</i> , and <i>Cassytha</i> are the most frequent in Indian forests.                                          |

The stratification system follows the framework described by Richards (1952, pp. 17–35) for tropical forests and by Whitmore (1984, pp. 30–35) for South and South-east Asian forests. The separation of IVa and IVb, and of Epi. and Par. as distinct entries, is characteristic of the silvicultural tradition of the Indian Forest Service, in which grasses signal degradation potential and epiphyte load signals stand maturity.

### A.8.2 Leaf habit (E / D)

| Code | Term      | Definition                                                                                                                                      |
|------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| E    | Evergreen | The species retains its leaves throughout the year; leaf fall and replacement are continuous rather than seasonal, so the canopy is never bare. |
| D    | Deciduous | The species sheds all or most of its leaves for a period each year, typically during the hot dry season in tropical India.                      |

These categories originate with Schimper (1903), who used them as primary physiognomic criteria for classifying vegetation formations. Raunkiaer (1934) formalised the distinction in his life-form system. In Champion & Seth, E and D appear against individual species in the floristic listings and are also used at Group level — the primary diagnostic criterion separating Tropical Wet Evergreen (Group 1) from Tropical Moist Deciduous (Group 3) is precisely the proportion of E and D species in the top canopy.

### A.8.3 Abundance scale

The abundance codes record the surveyor's field impression of how often and in what quantity a species is encountered within a stand. They are not measurements of density or cover; they are ordinal visual estimates. The scale is a variant of that introduced by Drude (1896), anglicised for use in British Empire vegetation survey by Tansley & Chipp (1926), and documented in its modern form by Kent (2012, Chapter 4).

| Code | Term             | Definition                                                                                                                                                                                                                      |
|------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (va) | Very abundant    | The species is met with in very large numbers throughout the stand; it forms a conspicuous mass of the vegetation and is near-dominant in its stratum.                                                                          |
| (a)  | Abundant         | Present in large numbers across most of the stand; a major component of the stratum it occupies, but not overwhelming other species.                                                                                            |
| (la) | Locally abundant | Abundant within restricted patches of the stand — dense where it occurs, but confined to particular localities such as a valley bottom, stream margin, or specific soil type, and absent or scarce elsewhere in the same stand. |
| (f)  | Frequent         | Encountered repeatedly and regularly as one moves through the stand; never surprising to find, but not massed.                                                                                                                  |
| (c)  | Common           | Widespread in the type across its geographic range; regularly present but not necessarily at high density at any one site.                                                                                                      |
| (lc) | Locally common   | Common within restricted patches of the stand but confined to particular localities and absent or scarce elsewhere.                                                                                                             |
| (o)  | Occasional       | Present but not regularly encountered; found as scattered individuals or in patches such that one could move through part of the stand without meeting it.                                                                      |
| (r)  | Rare             | Found only as isolated individuals or in very few spots within the stand; easily missed.                                                                                                                                        |

The "locally" modifier (la, lc) records that a species is **patchy rather than diffuse** — concentrated where site conditions favour it (a particular substrate, moisture level, or canopy gap) but absent from the rest of the stand. This is ecologically informative: a species marked (la) often signals an edaphic or microtopographic control within an otherwise uniform stand.

Because the scale is ordinal and visual, it is calibrated relative to the stand rather than to an absolute standard. A species marked (f) in a species-poor thorn forest and one marked (f) in a species-rich wet evergreen forest are not at the same stem density; the label records relative prominence within the stand as perceived by the surveyor. This limitation is well documented in Kent (2012) and should be kept in mind when the abundance codes are used analytically.

## A.9 References

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